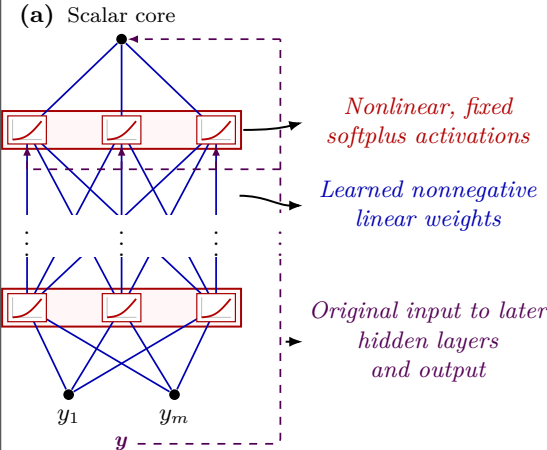
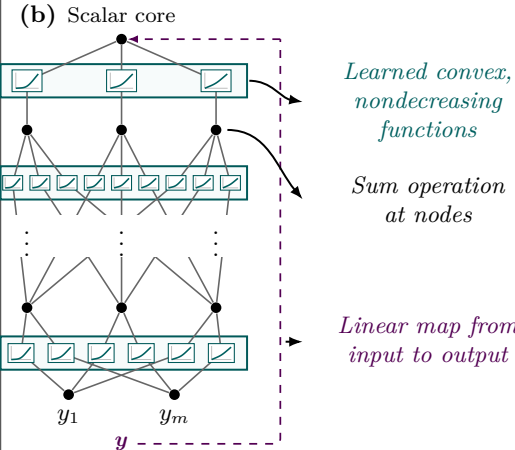


Core	Monotone ICNN	Monotone integrated-hat ICKAN
Layer operation	Learned nonnegative linear weights + biases $\longrightarrow$ fixed softplus at hidden nodes	Learned scalar connection functions (convex and nondecreasing) $\longrightarrow$ sum at nodes + biases
Multilayer core	(a) Scalar core  Nonlinear, fixed softplus activations Learned nonnegative linear weights Original input to later hidden layers and output	(b) Scalar core  Learned convex, nondecreasing functions Sum operation at nodes Linear map from input to output
	— Learned linear weights Fixed softplus Learned spline functions --- Input maps All linear weights are nonnegative; biases are unrestricted.	